

Load-Dependent Sacroiliac Joint Micromotion in a Large Pelvic Cohort: Directional Re-routing, Between-Subject Variability and Scaling

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Abstract

The human sacroiliac joint (SIJ) functions at the biomechanical interface between bipedal weight-bearing stability and cephalopelvic obstetric clearance. Under the paradigm of form and force closure, axial locomotor loads engage self-bracing nutation, but how multi-axial parturition-motivated forces re-route joint kinematics, the degree of sex-related mechanical compliance, and the relative contributions of skeletal geometry versus bone tissue material heterogeneity remain unresolved across population cohorts. We evaluated a precomputed finite-element cohort of 278 human pelves (150 female, 128 male; age 16–91 years) parameterized via an implicit subject-specific continuous displacement mapping on a shared reference template. Five standardized static loading regimes were analyzed with the superior S1 facet constrained via a Dirichlet penalty: bilateral standing (SP2leg; +400 N vertical ground reaction on each acetabular notch, 800 N total), unilateral standing (SP1leg; +800 N on the right acetabulum), and three parturition-motivated multi-axial expansion configurations (LAB1–LAB3; ± 400 N internal ring compression, ischial distraction, and outlet distraction proxies). Kinematics were resolved in a local joint coordinate triad, tracking bilateral means of 3D resultant norms ($|\boldsymbol{\theta}|$, $|\mathbf{d}|$) and absolute component magnitudes. We tested directional contrasts relative to bilateral standing, assessed sex-related differences through nested mul-

tivariable linear regressions (M0: sex and age; M1: +pelvic volume; M2: +morphometric triad) with heteroskedasticity-consistent standard errors (HC3) and variance inflation factor screening, evaluated variance retention ratios ($V_{\text{shape}}/V_{\text{full}}$, $V_{\text{material}}/V_{\text{full}}$) with paired subject-level diagnostics, and explored allometric power-law scaling with false discovery rate control ($q < 0.05$). Stance regimes engaged physiological nutation (median 1.06° in SP2leg, 1.16° in SP1leg). Unilateral stance imposed the highest kinematic demand ($|\theta| = 2.52^\circ$, $|\mathbf{d}| = 1.76$ mm), driving bilateral translation asymmetry from 0.21 mm in SP2leg to 2.01 mm. Under parturition proxies, sagittal nutation was suppressed (0.41° – 0.43° in LAB1/LAB2), and displacement shifted from vertical craniocaudal shear ($\Delta\text{CC} = -0.59$ to -0.90 mm) to transverse outward distraction ($\Delta\text{ML} = +0.09$ to $+0.15$ mm). In multivariable models, female pelves exhibited a significant translational surplus across labor regimes ($+0.19$ to $+0.23$ mm, $p \leq 0.020$), which attenuated by approximately 38% upon adjusting for volume and morphometric dimensions; transverse compliance was positively associated with subpubic arch angle ($r = 0.52$, $p < 10^{-19}$). Variance decomposition demonstrated geometric dominance: shape-only models retained 98.7%–102.1% (median 100.5%) of full-model variance ($R^2 \geq 0.994$, $\text{RMSE} \leq 0.031$ mm), whereas bone mineral density heterogeneity accounted for $< 0.32\%$ (median 0.033%). Negative allometric volume scaling was strongest in unilateral stance ($\beta_{\text{male}} = -0.57$, $\beta_{\text{female}} = -0.43$, $q \leq 0.004$). Under standardized static loading, SIJ micromotion displays marked directional re-routing: locomotor loading reinforces self-bracing nutation, whereas parturition-motivated regimes direct compliance into transverse ring expansion. Between-subject kinematic variation is governed predominantly by pelvic geometry rather than bone mineralization, clarifying the passive mechanical compromise between bipedal stability and pelvic compliance.

Keywords: sacroiliac joint; pelvic functional anatomy; finite-element analysis; nutation; form and force closure; sex-related differences; pelvic architecture; variance channels

1 Introduction

The human sacroiliac joint (SIJ) represents an important mechanical compromise in hominin evolution: it must provide stable transmission of ground reaction forces during bipedal locomotion while simultaneously accommodating the compliance required for cephalopelvic clearance during parturition (Washburn, 1960; Ruff, 2017; Pavličev et al., 2020; Whitcome et al., 2017). Biomechanically, this dual demand is governed by the principles of *form closure* and *force closure* (Vleeming et al., 2012; Snijders et al., 1993). Form closure is conferred by the interlocking wedge-like architecture of the sacrum within the pelvic ring, characterized by biconcave–biconvex

articular congruency, macroscopic chondral ridges, and high frictional resistance (Vleeming et al., 2012; Kiapour et al., 2019). Force closure is provided dynamically by surrounding musculature and passively by an extensive ligamentous complex, including the anterior, interosseous, and dorsal sacroiliac ligaments, as well as the sacrotuberous and sacrospinous ligaments (Snijders et al., 1993; Hammer et al., 2013, 2019).

Under upright axial gravity loading, the functional stability of the pelvic ring relies on *nutation*—the anterior-inferior rotation of the sacral promontory relative to the ilia (Vleeming et al., 2012; Snijders et al., 1993; Keizer et al., 2019). Nutation increases tension across the primary posterior ligamentous stabilizers, drawing the auricular surfaces into tight apposition and self-bracing the articulation against vertical shear (Snijders et al., 1993; Vleeming et al., 2012). During cyclical gait, unilateral single-leg support (the single-stance phase) amplifies this demand: ground reaction forces are transmitted through a single acetabulum while body weight acts eccentrically on the contralateral side, generating pelvic tilt and asymmetric shear across the pelvic arch (Bergmann et al., 2001; Kutzner et al., 2016; Hammer et al., 2013).

In contrast to upright locomotor support, parturition imposes an opposing mechanical demand: circumferential and transverse expansion of the pelvic cavity (Pavličev et al., 2020; Mora et al., 2022; House et al., 2022). As the fetal head traverses the inlet, midplane, and outlet, the surrounding pelvic ring is subjected to internal outward pressures and soft-tissue tensions that challenge sacroiliac joint congruency (Mora et al., 2022; Fuchs et al., 2013; Roque de Almeida et al., 2024). While pregnancy-related hormones such as relaxin contribute to systemic ligamentous laxity in vivo, the underlying bony and articular geometry of the pelvic ring fundamentally determines the passive kinematic pathways through which external expansion loads are absorbed (Ruff, 2017; Betti and Manica, 2020; Weber et al., 2019; Weiss, 2020). Although in vivo experimental studies consistently describe the SIJ as a low-motion joint (sub-millimetric translations and rotations rarely exceeding a few degrees; (Goode et al., 2008; Meijer et al., 2014; Keizer et al., 2019; Noë et al., 2020)), they cannot isolate how differing loading directions selectively re-route 3D joint displacement vectors across diverse pelvic morphologies.

Finite-element (FE) analysis provides a valuable methodology to resolve these joint-level kinematics under standardized boundary conditions (Anderson et al., 2020; Yang et al., 2020; Song et al., 2021; Calixte et al., 2022). However, two fundamental questions remain unresolved in pelvic functional anatomy. First, how are standardized 3D force vectors partitioned into local joint degrees of freedom? Specifically, does simulated obstetric loading simply attenuate over-

all micromotion, or does it execute a qualitative *directional re-routing* from sagittal nutation into transverse ring expansion? Second, what is the primary determinant of between-subject kinematic variability in the human population? Population variance in joint micromotion could plausibly originate from two distinct mechanistic “channels”: macro-architectural geometric variation (pelvic proportions, inlet breadth, subpubic angle, and articular orientation; (Ruff, 2017; Betti and Manica, 2020; Weber et al., 2019)) or tissue-level material heterogeneity (heterogeneous bone mineralization and apparent density; (Carter and Hayes, 1977; Keller, 1994; Morgan et al., 2003; Frost, 1987; Huiskes et al., 1987)). Establishing whether anatomy or bone tissue quality governs SIJ kinematics is important for evolutionary morphology, biomechanical assessment of peripartum pelvic girdle pain, and orthopedic surgical fixations (Hammer et al., 2019; Kiapour et al., 2019).

To resolve these questions, we investigated a standardized finite-element repository comprising 278 human pelves (150 female, 128 male; age 16–91 years) parameterized on a common reference template via an implicit continuous displacement mapping (Henyš and Kuchař, 2025). The dataset features fully coupled, shape-only, and material-only simulation variants evaluated across five standardized static loading regimes spanning bipedal locomotion and multi-axial parturition proxies. We tested four anatomically grounded hypotheses:

Hypotheses

1. **H1 (Load-Dependence and Stance Asymmetry):** Bilateral and unilateral upright stance will engage physiological self-bracing nutation (θ_{nut}), with unilateral stance generating substantial bilateral kinematic asymmetry due to asymmetric acetabular support. Conversely, parturition-motivated mechanical proxies will de-emphasize sagittal nutation and vertical shear in favor of transverse compliance.
2. **H2 (Directional Contrasts in Kinematic State Space):** Parturition-motivated configurations will recruit directional displacements distinct from locomotor stance, shifting the dominant motion from craniocaudal shear (ΔCC) to transverse outward distraction (ΔML) and sagittal translation.
3. **H3 (Sex-Related Differences and Morphometric Associations):** Female pelves will exhibit greater translational compliance under parturition-motivated configurations after adjusting for age and pelvic size, with this difference being attenuated when conditioning

on transverse pelvic architecture (particularly subpubic arch angle and biischadic width).

4. H4 (Relative Contributions of Shape versus Material Heterogeneity): Between-subject kinematic variance across the population will be governed primarily by macroscopic skeletal geometry rather than bone mineral density heterogeneity.

As an exploratory secondary objective, we also evaluated allometric scaling relationships between pelvic volume and SIJ micromotion across all load configurations.

2 Materials and methods

2.1 Cohort characteristics and ethical governance

We investigated a standardized finite-element pelvic repository and two matched auxiliary variant datasets developed to isolate structural shape from bone tissue material heterogeneity (Henyš and Kuchař, 2025). The cohort comprised 278 complete human pelvic specimens (150 female, 128 male) spanning an adult lifespan of 16 to 91 years (mean $\pm$ SD: 53.9 ± 19.3 years). All models were generated from high-resolution clinical computed tomography (CT) scans curated within the open BoneDat anatomical repository (Henyš and Kuchař, 2025). Use of these anonymized imaging data was approved by the Ethics Committee of the University Hospital Hradec Králové (reference 202411IO3P) with a waiver of informed consent.

Eight standardized linear and angular morphometric landmarks were extracted for each pelvis (Table 1): anteroposterior (AP) inlet diameter, biacetabular width, biischadic width, bituberous width, iliopectineal eminence width, pelvic inlet transverse (PIT) diameter, sacral width, and subpubic arch angle (see Fig. 1A and Appendix A for landmark definitions of the primary morphometric triad). Morphometric completeness, demographic metadata, global volume metrics, and SIJ kinematic endpoints exhibited zero missing values across the 278 subjects.

Table 1: Cohort demographic and morphometric characteristics stratified by sex. Values represent mean $\pm$ SD.

Variable	Overall	Female	Male	Missing
N	278	150	128	0
Age (years)	53.90 $\pm$ 19.33	53.36 $\pm$ 19.74	54.52 $\pm$ 18.91	0
Inlet AP diameter (mm)	117.04 $\pm$ 10.52	121.04 $\pm$ 10.02	112.34 $\pm$ 9.07	0
Biacetabular width (mm)	125.81 $\pm$ 8.53	129.08 $\pm$ 8.21	121.98 $\pm$ 7.22	0
Biischiadic width (mm)	104.27 $\pm$ 9.33	109.87 $\pm$ 7.13	97.71 $\pm$ 7.06	0
Bituberous width (mm)	132.24 $\pm$ 9.15	136.46 $\pm$ 8.29	127.30 $\pm$ 7.49	0
Iliopectineal eminence width (mm)	102.38 $\pm$ 7.75	106.20 $\pm$ 7.02	97.90 $\pm$ 5.96	0
Pelvic inlet transverse (mm)	131.58 $\pm$ 8.88	134.85 $\pm$ 8.59	127.75 $\pm$ 7.61	0
Sacral width (mm)	117.08 $\pm$ 6.08	116.75 $\pm$ 5.67	117.47 $\pm$ 6.54	0
Subpubic angle (deg)	78.12 $\pm$ 10.12	85.80 $\pm$ 6.14	69.11 $\pm$ 5.26	0
Global scale factor (-)	0.96 $\pm$ 0.06	0.92 $\pm$ 0.05	1.00 $\pm$ 0.05	0
Total pelvic volume (cm ³)	823.58 $\pm$ 149.32	736.79 $\pm$ 105.99	925.28 $\pm$ 126.99	0
Total pelvic surface (cm ²)	1568.63 $\pm$ 169.26	1479.09 $\pm$ 133.64	1673.57 $\pm$ 144.80	0

2.2 Finite-element modeling and geometry parameterization

Simulations were formulated under a 3D small-strain continuum elastostatic framework posed on a shared, high-fidelity reference pelvic template (Bonet and Wood, 1997; Marsden and Hughes, 1983) and solved using FEniCSx/DOLFINx (v0.10) with PETSc linear algebraic solvers. Subject-specific anatomy was parameterized through a smooth, invertible displacement vector field $\phi(\mathbf{X})$, mapping reference spatial coordinates $\mathbf{X}$ to individual anatomy $\mathbf{x} = \mathbf{X} + \phi(\mathbf{X})$ via radial basis function interpolation (Henyš and Kuchař, 2025). This implicit mapping avoids mesh discretization discrepancies across subjects while preserving exact individual curvature, cortical boundaries, and joint orientations. Conforming, continuous meshes were maintained across bone–cartilage interfaces at both sacroiliac joints and the pubic symphysis.

Bony material properties were assigned by mapping voxel-level CT attenuation values into apparent bone mineral density and computing local elastic moduli via established power-law formulations (Carter and Hayes, 1977; Keller, 1994; Morgan et al., 2003; Frost, 1987; Huiskes et al., 1987):

$$E = \begin{cases} 2398 \text{ MPa}, & \rho_{\text{ash}} \leq 0.486 \\ 10200 \rho_{\text{ash}}^2 \text{ MPa}, & \rho_{\text{ash}} > 0.486 \end{cases}, \quad \rho_{\text{ash}} = \frac{\rho + 0.09}{1.14}, \quad (1)$$

with an isotropic Poisson’s ratio $\nu = 0.30$. The bilateral sacroiliac joint cartilages and the fibrocartilaginous pubic symphysis were modeled with nominal linear-elastic properties ($E = 1.0 \text{ MPa}$, $\nu = 0.45$), consistent with established pelvic finite-element parameterizations (Ander-

son et al., 2020; Yang et al., 2020; Calixte et al., 2022). The multi-ligamentous pelvic apparatus was represented explicitly via discrete spring elements with non-linear tensile stiffness and nominal pretension, incorporating anterior sacroiliac, posterior/dorsal sacroiliac, interosseous sacroiliac, sacrotuberous, sacrospinous, and pubic ligament complexes (Hammer et al., 2013, 2019).

2.3 Standardized physiological loading regimes

In all configurations, the pelvic ring was constrained at the superior S1 sacral facet endplate via a Dirichlet boundary penalty condition ($\gamma = 1.0 \times 10^6$). Five standardized static load configurations were evaluated (Table 2; Fig. 1B–D):

1. **Two-Leg Stance (SP2leg):** Symmetrical bipedal stance under bilateral vertical upward ground reaction forces of +400 N applied to each acetabular notch (resultant +800 N craniocaudal force reacting against the fixed sacrum).
2. **Single-Leg Stance (SP1leg):** Unilateral stance phase under a +800 N vertical upward ground reaction force applied to the right acetabular notch (resultant +800 N craniocaudal force), simulating single-limb support with an unsupported contralateral hemi-pelvis.
3. **Labor Phase 1 (LAB1 - Internal Ring Compression/Shear Proxy):** Mediolateral force pair (± 400 N; +400 N on left, -400 N on right internal pelvic ring contact surfaces), modeling inward/transverse head engagement during early fetal descent.
4. **Labor Phase 2 (LAB2 - Ischial Distraction Proxy):** Bilateral mediolateral distraction under ± 400 N outward forces applied to the left and right ischial tuberosities (+400 N left, -400 N right), representing midplane transverse dilation.
5. **Labor Phase 3 (LAB3 - Outlet AP Distraction Proxy):** Anteroposterior outlet distraction under ± 400 N opposing forces acting between the anterior pubic insertion (-400 N anteriorly) and the posterior sacrococcygeal complex (+400 N posteriorly), simulating sagittal outlet expansion during fetal expulsion.

Forces were applied as standardized surface tractions integrated over designated anatomical surface regions, scaled inversely by surface area to ensure identical resultant force magnitudes across all individual pelvic morphologies (see Appendix A for complete mathematical details of the boundary value problem and traction integrals).

Table 2: Standardized static load cases spanning locomotor and parturition-motivated mechanical regimes.

Load case	Mechanical proxy	Applied total forces	Resultant
SP2leg	Bilateral standing	[0,0,+400] N on right AC notch and [0,0,+400] N on left AC notch	800 N cranio-caudal
SP1leg	Unilateral standing	[0,0,+800] N on right AC notch	800 N cranio-caudal
LAB1	Mediolateral ring compression proxy	[+400,0,0] N on ring_contact_left and [-400,0,0] N on ring_contact_right	0 N net; bilateral ML pair
LAB2	Ischial loading / SIJ distraction proxy	[+400,0,0] N on left_ischium_tuber and [-400,0,0] N on right_ischium_tuber	0 N net; bilateral ML pair
LAB3	Outlet AP distraction proxy	[0,-400,0] N on pubis_ins and [0,+400,0] N on SCJ	0 N net; AP pair

2.4 SIJ kinematics and local joint coordinate triad

Relative motion of the ilium with respect to the sacrum was extracted from the deformed nodal fields of the bilateral articulation surfaces using trimmed rigid registration (Kabsch algorithm) on the anatomical regions of interest (Henyš and Kuchař, 2025). A complete mathematical specification of pelvic coordinate frame construction, affine sacrum drift removal, trimmed Kabsch registration, and Cardan angle extraction is detailed in Appendix A. Kinematics were projected onto an anatomical joint coordinate triad following a Cardan rotation sequence (XYZ order; Fig. 1D):

- **Nutation** ($\theta_{\text{nut}} = \theta_x$): Sagittal rotation around the joint transverse axis (positive = anterior sacral tilt/nutation, negative = posterior sacral tilt/counternutation).
- **Abduction** (θ_y): Frontal tilt around the anteroposterior axis.
- **Axial Rotation** (θ_z): Transverse twist around the craniocaudal axis.
- **Translations** (dx , dy , dz): 3D translation components along the mediolateral (ML; dx), anteroposterior (AP; dy), and craniocaudal (CC; dz) axes.

Scalar kinematic endpoints were defined as the 3D resultant rotation magnitude,

$$|\boldsymbol{\theta}| = \frac{1}{2} (\|\boldsymbol{\theta}_{\text{left}}\| + \|\boldsymbol{\theta}_{\text{right}}\|), \quad (2)$$

and 3D resultant translation magnitude,

$$|\mathbf{d}| = \frac{1}{2} (\|\mathbf{d}_{\text{left}}\| + \|\mathbf{d}_{\text{right}}\|), \quad (3)$$

evaluated as the bilateral mean across left and right SIJs. Component motions reported in summary tables and statistical contrasts represent bilateral means of absolute component magnitudes ($0.5[|t_{L,k}| + |t_{R,k}|]$ for $k \in \{\text{AP}, \text{ML}, \text{CC}\}$ and nutation $|\theta_x|$), preventing sign-cancellation

across bilateral mirror symmetry. Bilateral asymmetry was quantified as the absolute inter-side difference: $\Delta_{\text{asym}}|\mathbf{d}| = |d_{\text{left}} - d_{\text{right}}|$.

2.5 Variance-channel decomposition and paired diagnostics

To decouple macroscopic geometry from tissue material properties, each subject's response was evaluated across three matched simulation repositories:

1. **Full Model:** Subject-specific mapped geometry $\phi(\mathbf{X})$ and subject-specific bone density mapping $E(\mathbf{x})$.
2. **Shape-Only Model:** Subject-specific mapped geometry $\phi(\mathbf{X})$ evaluated with homogeneous, population-median reference bone material properties.
3. **Material-Only Model:** Subject-specific bone density mapping $E(\mathbf{x})$ applied to the unmapped reference template geometry ($\phi = \mathbf{0}$).

Importantly, "material-only" isolates cortical and trabecular bone mineral density (modulus) heterogeneity, while cartilage and ligament properties remain nominal. For each load case and kinematic metric, variance retention ratios were computed as $V_{\text{shape}}/V_{\text{full}}$ and $V_{\text{material}}/V_{\text{full}}$, with 95% confidence intervals estimated via 2,000 bootstrap iterations (Efron and Tibshirani, 1994). Because these ratios compare sample variances across separate datasets, they represent variance retention ratios rather than an orthogonal additive decomposition. To assess subject-level agreement between the Full and Shape-Only models, we evaluated paired diagnostics including the coefficient of determination against the 1:1 identity line (R^2), Pearson and Spearman correlation coefficients, root-mean-square error (RMSE), and mean absolute error (MAE).

2.6 Statistical analysis

Continuous outcomes were summarized as medians and interquartile ranges [IQR]. Directional re-routing was tested by computing subject-wise differences against baseline bilateral stance ($\Delta = \text{Outcome}_{\text{load}} - \text{Outcome}_{\text{SP2leg}}$) and evaluating shifts with Wilcoxon signed-rank tests (Wilcoxon, 1945).

To evaluate sex-related differences and morphometric associations under parturition loads (H3), we fitted nested ordinary least squares (OLS) regressions with HC3 heteroskedasticity-

consistent standard errors (White, 1980; MacKinnon and White, 1985):

$$\text{Model 0 (M0): } Y = \beta_0 + \beta_{\text{sex}} \cdot \text{Sex} + \beta_{\text{age}} \cdot \text{Age}_z + \epsilon, \quad (4)$$

$$\text{Model 1 (M1): } Y = \beta_0 + \beta_{\text{sex}} \cdot \text{Sex} + \beta_{\text{age}} \cdot \text{Age}_z + \beta_{\text{vol}} \cdot \log(\text{Vol})_z + \epsilon, \quad (5)$$

$$\begin{aligned} \text{Model 2 (M2): } Y = & \beta_0 + \beta_{\text{sex}} \cdot \text{Sex} + \beta_{\text{age}} \cdot \text{Age}_z + \beta_{\text{vol}} \cdot \log(\text{Vol})_z \\ & + \beta_{\text{AP}} \cdot \text{AP}_z + \beta_{\text{ischiadic}} \cdot \text{Biischiadic}_z + \beta_{\text{subpubic}} \cdot \text{Subpubic}_z + \epsilon, \end{aligned} \quad (6)$$

where continuous predictors were standardized (z -scores). The morphometric triad (AP diameter, biischiadic width, and subpubic angle; defined geometrically in Appendix A) was chosen *a priori* to capture inlet depth, transverse outlet span, and anterior arch opening. Multicollinearity among predictors was screened using Variance Inflation Factors (VIF). Progressive attenuation of β_{sex} across $\text{M0} \rightarrow \text{M1} \rightarrow \text{M2}$ evaluated the proportion of sex-related mobility attributable to size and architectural dimorphism. In addition, bivariate associations between kinematic compliance and individual morphometric landmarks were examined using Spearman rank correlations.

Repeated measures across all five load cases were analyzed using cluster-robust OLS clustered at the subject level. Secondary allometric size-scaling relationships were evaluated using log–log power-law regressions:

$$\log(Y) = \alpha + \beta_{\text{vol}} \cdot \log(\text{Total Volume}) + \beta_{\text{sex}} \cdot \text{Sex} + \beta_{\text{int}} \cdot [\log(\text{Total Volume}) \times \text{Sex}] + \mathbf{Z}\boldsymbol{\gamma} + \epsilon, \quad (7)$$

extracting sex-stratified allometric scaling exponents (β_{male} and β_{female}) with Wald tests for non-zero slopes. Multiplicity across hypothesis tests was controlled using the Benjamini–Hochberg false discovery rate (BH-FDR, $q < 0.05$) (Benjamini and Hochberg, 1995).

3 Results

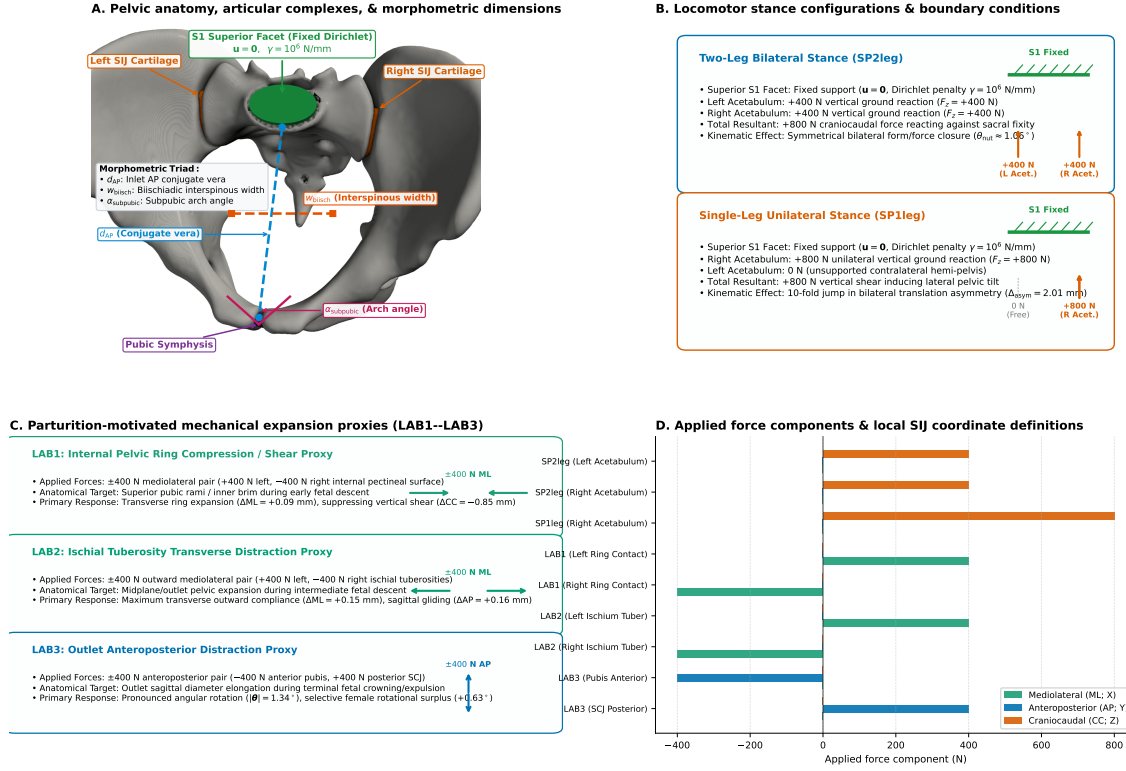


Figure 1: Pelvic anatomical framework, morphometric dimensions, boundary conditions, and standardized loading regimes. (A) 3D finite-element rendering of the human pelvis showing the bilateral sacroiliac joint (SIJ) articular cartilages (orange), the superior S1 sacral facet constrained via a Dirichlet penalty ($\mathbf{u} = \mathbf{0}$, $\gamma = 10^6$ N/mm; green), and the anterior fibrocartilaginous pubic symphysis disc (purple). Overlays mark the primary morphometric triad: anteroposterior inlet diameter (conjugate vera d_{AP} , blue), biischadic interspinous width (w_{biisch} , amber), and subpubic arch angle ($\alpha_{subpubic}$, magenta). (B) Locomotor stance configurations and boundary conditions: symmetrical two-leg stance (SP2leg; +400 N per acetabular notch, +800 N total vertical resultant reacting against the constrained sacral base) and single-leg stance (SP1leg; +800 N unilateral vertical ground reaction on the right acetabulum with unsupported contralateral hemi-pelvis). (C) Parturition-motivated mechanical expansion proxies: LAB1 (internal pelvic ring compression/shear proxy; ± 400 N mediolateral pair on pectineal surfaces), LAB2 (ischial tuberosity transverse distraction proxy; ± 400 N outward mediolateral pair), and LAB3 (outlet anteroposterior distraction proxy; -400 N anterior pubis, +400 N posterior sacrococcygeal joint). (D) Applied resultant force components across anatomical axes: mediolateral (ML, green), anteroposterior (AP, blue), and craniocaudal (CC, vermilion). Mathematical formulations for the continuum boundary value problem, landmark extractions, rigid kinematic registration, and local Cardan angle extractions are detailed in Appendix A.

3.1 Cohort completeness and morphometric sexual dimorphism

The analyzed cohort comprised 278 human pelvic models (150 female, 128 male; age 53.9 ± 19.3 years, range 16–91). All 278 datasets exhibited complete records across demographic metadata, eight 3D anatomical dimensions, and finite-element kinematic outcomes (Table 1). Morpho-

metric comparisons confirmed characteristic pelvic sexual dimorphism: female pelves exhibited significantly larger mean anteroposterior inlet diameter (121.04 ± 10.02 vs. 112.34 ± 9.07 mm), biischiadic outlet width (109.87 ± 7.13 vs. 97.71 ± 7.06 mm), and subpubic arch angle ($85.80 \pm 6.14^\circ$ vs. $69.11 \pm 5.26^\circ$), despite having lower total pelvic bone volume (736.79 ± 105.99 vs. 925.28 ± 126.99 cm³) and lower global scale factors (0.92 ± 0.05 vs. 1.00 ± 0.05) than male pelves.

3.2 Functional load-dependence: locomotor self-bracing versus asymmetric shear (H1)

Sacroiliac joint kinematics varied markedly across loading regimes (Table 3; Fig. 2). Bipedal standing (SP2leg) engaged physiological nutation (median $\theta_{\text{nut}} = 1.062^\circ$ [IQR 0.653, 1.522]), maintaining low bilateral translation asymmetry (0.215 mm [0.093, 0.347]) and moderate resultant motion ($|\theta| = 1.692^\circ$ [1.211, 2.451]; $|\mathbf{d}| = 1.299$ mm [0.971, 1.625]).

Transition to single-leg stance (SP1leg) produced the highest kinematic demand among the evaluated states, driving resultant rotation to 2.516° [1.981, 3.137] and translation to 1.755 mm [1.458, 2.053]. Unilateral acetabular support induced a marked increase in bilateral translation asymmetry, rising from 0.215 mm in SP2leg to 2.013 mm [1.423, 2.548] in SP1leg (Fig. 2D). While sagittal nutation remained elevated (1.160° [0.819, 1.619]), unilateral vertical support imposed substantial frontal pelvic tilt (abduction $\theta_y = 1.582^\circ$ [1.195, 1.977]), generating differential shear across the pelvic arch.

Table 3: SIJ micromotion summary by load case across 278 subjects. Values represent median [interquartile range].

Load case	N	Rotation magnitude (deg)	Translation magnitude (mm)	AP translation component (mm)	ML translation component (mm)	CC translation component (mm)
SP2leg	278	1.692 [1.211, 2.451]	1.299 [0.971, 1.625]	0.272 [0.158, 0.433]	0.318 [0.160, 0.574]	1.131 [0.855, 1.459]
SP1leg	278	2.516 [1.981, 3.137]	1.755 [1.458, 2.053]	0.431 [0.299, 0.572]	0.341 [0.200, 0.550]	1.527 [1.262, 1.775]
LAB1	278	1.393 [0.908, 2.060]	0.665 [0.450, 0.923]	0.268 [0.169, 0.424]	0.410 [0.228, 0.678]	0.263 [0.163, 0.410]
LAB2	278	1.433 [1.011, 2.069]	0.764 [0.536, 1.048]	0.413 [0.216, 0.628]	0.476 [0.284, 0.738]	0.236 [0.138, 0.377]
LAB3	278	1.769 [1.169, 2.591]	0.790 [0.568, 1.052]	0.262 [0.143, 0.473]	0.274 [0.138, 0.511]	0.534 [0.325, 0.749]

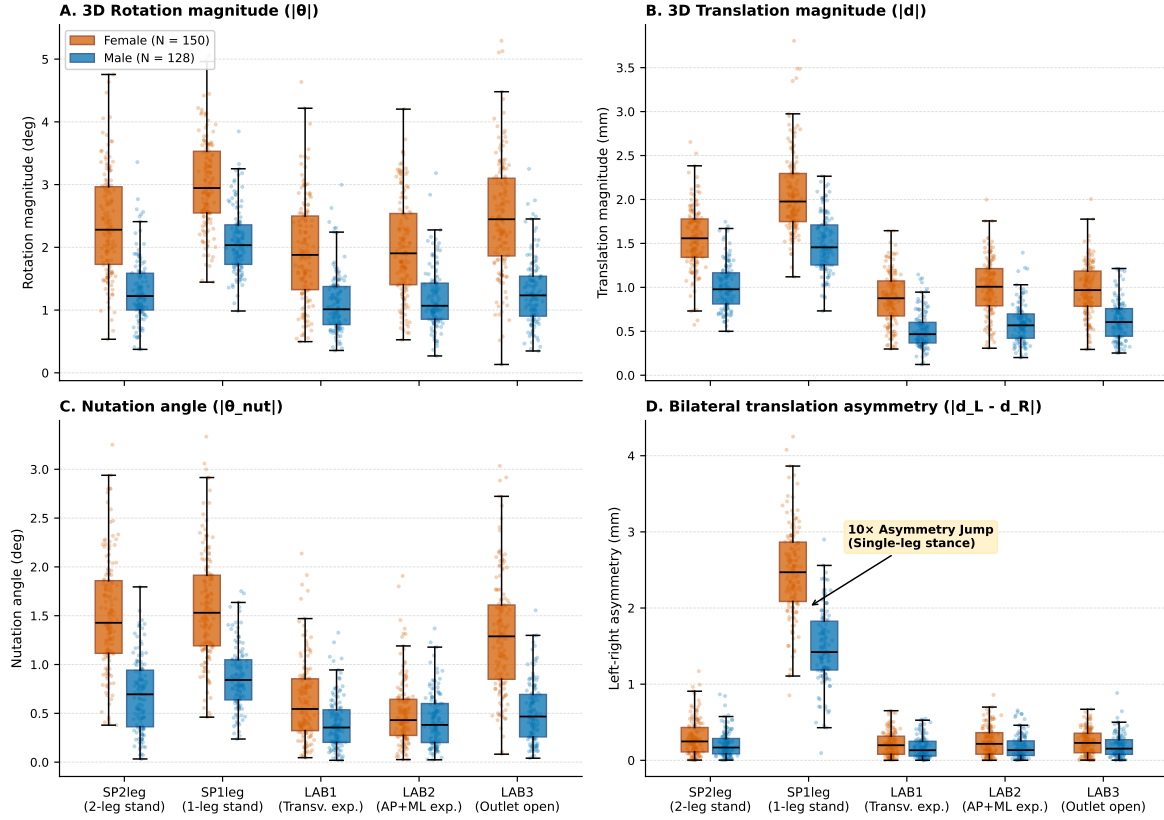


Figure 2: **Primary SIJ kinematic distributions across locomotor and parturition load regimes.** Raincloud distribution dashboard showing: (A) 3D resultant rotation magnitude ($|\theta|$, deg), (B) 3D resultant translation magnitude ($|d|$, mm), (C) sagittal nutation angle (θ_{nut} , deg), and (D) bilateral translation asymmetry ($|d_{\text{left}} - d_{\text{right}}|$, mm). Boxplots depict medians and interquartile ranges; dashed reference lines indicate quiet stance baselines (SP2leg). Note the reduction in nutation under simulated labor (C) and the marked elevation in bilateral asymmetry during single-leg stance (D).

3.3 Directional kinematic contrasts in 3D state space (H2)

Under parturition-motivated configurations (LAB1–LAB3), SIJ kinematics exhibited directional re-routing rather than uniform scaling (Table 3; Fig. 3). In contrast to stance, sagittal nutation decreased under internal ring expansion, dropping to 0.432° [0.257, 0.731] in LAB1 and 0.412° [0.247, 0.630] in LAB2 (Fig. 2C).

Decomposition of within-subject displacement differences relative to SP2leg demonstrated directional anisotropy (Fig. 3A; Supplementary Table S1). Single-leg stance selectively increased vertical shear ($\Delta\text{CC} = +0.394$ mm, 95% CI 0.347 to 0.434, $p < 10^{-40}$) and anteroposterior translation ($\Delta\text{AP} = +0.072$ mm, 95% CI 0.045 to 0.096, $p < 10^{-20}$), while mediolateral displacement showed no significant shift ($\Delta\text{ML} = +0.005$ mm, $p = 0.095$). Conversely, internal ring compression (LAB1) produced an abrupt reduction in craniocaudal shear ($\Delta\text{CC} = -0.850$ mm,

95% CI -0.881 to -0.811) while channeling displacement into transverse outward distraction ($\Delta ML = +0.086$ mm, 95% CI 0.077 to 0.095 , $p < 10^{-30}$). Circumferential dilation (LAB2) similarly suppressed vertical shear ($\Delta CC = -0.903$ mm, 95% CI -0.931 to -0.853) while augmenting both transverse distraction ($\Delta ML = +0.145$ mm, 95% CI 0.132 to 0.160) and sagittal gliding ($\Delta AP = +0.163$ mm, 95% CI 0.102 to 0.213).

Projecting these responses into a 2D kinematic state space (ΔML vs. ΔCC ; Fig. 3B) highlights the directional contrast: locomotor stance contrasts populate the vertical shear axis, whereas parturition proxies diverge along the transverse distraction axis.

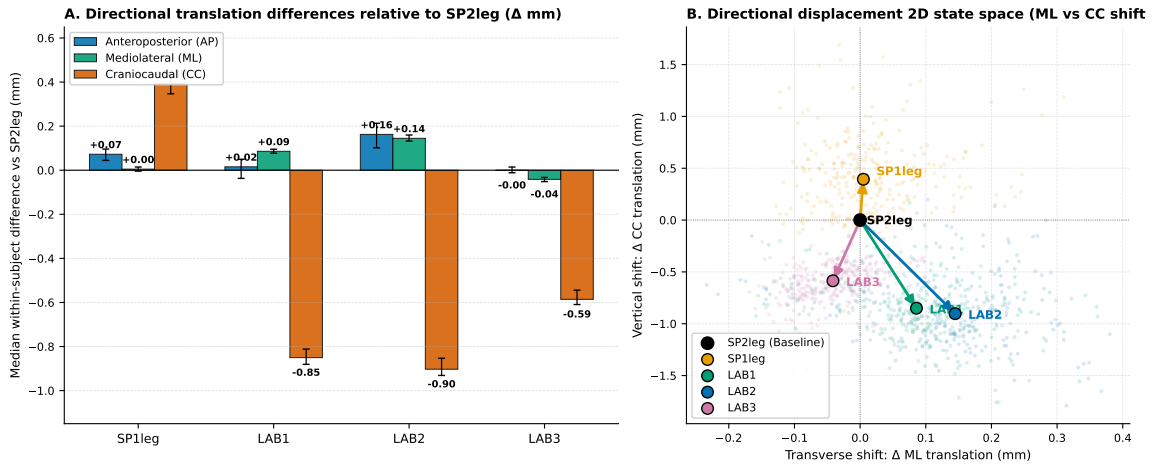


Figure 3: Directional translation differences and 2D state-space contrast. (A) Within-subject translation shifts ($\Delta = \text{Load} - \text{SP2leg}$) for anteroposterior (AP, blue), mediolateral (ML, green), and craniocaudal (CC, vermilion) axes. Error bars indicate bootstrap 95% confidence intervals; asterisks denote significant median shifts after BH-FDR correction ($q < 0.05$). (B) Kinematic state-space contrast mapping transverse shifts (ΔML) against vertical shear shifts (ΔCC), highlighting the functional divergence between locomotor vertical shear and obstetric transverse ring compliance.

3.4 Sex-related differences and morphometric associations (H3)

In unadjusted models conditioned only on age (Model 0; Eq. 4), female pelvis demonstrated greater translation magnitude under all parturition configurations: $+0.364$ mm in LAB1 (SE = 0.030 , $p < 10^{-30}$), $+0.414$ mm in LAB2 (SE = 0.033 , $p < 10^{-35}$), and $+0.344$ mm in LAB3 (SE = 0.031 , $p < 10^{-25}$). Adjusting for total pelvic bone volume (Model 1; Eq. 5) attenuated the female surplus to $+0.295$ mm in LAB1, $+0.329$ mm in LAB2, and $+0.279$ mm in LAB3 ($p < 10^{-9}$ for all).

Further adjustment for the morphometric triad (Model 2; Eq. 6: AP diameter, biischiadic width, and subpubic arch angle) reduced the conditional difference (female minus male) to

+0.224 mm in LAB1 (SE = 0.078, 95% CI 0.070 to 0.378, $p = 0.0042$), +0.193 mm in LAB2 (SE = 0.083, 95% CI 0.031 to 0.355, $p = 0.0196$), and +0.231 mm in LAB3 (SE = 0.080, 95% CI 0.075 to 0.388, $p = 0.0038$; Fig. 4A,B; Supplementary Table S3). Across Model 2, multicollinearity diagnostics confirmed acceptable variance inflation factors: female sex (VIF = 5.55), subpubic angle (VIF = 4.76), biischiadic width (VIF = 3.03), pelvic volume (VIF = 2.62), AP diameter (VIF = 1.59), and age (VIF = 1.55), all well below the standard threshold of 10. Rotational dimorphism was strongest in outlet distraction (LAB3: +0.627°, 95% CI 0.206 to 1.048, BH-FDR $p = 0.007$).

In pooled cluster-robust models across all load regimes, female sex retained significant positive associations for both rotation (+0.450°, $p = 0.020$) and translation (+0.254 mm, $p < 0.001$; Table 4A). Morphometric associations revealed that enhanced compliance is linked with anterior arch architecture: LAB1 translation correlated positively with the subpubic arch angle (Spearman $r = 0.52$, $p = 2.2 \times 10^{-20}$; Fig. 4C), indicating that wider anterior arch geometry facilitates outward compliance of the dorsal sacroiliac joints.

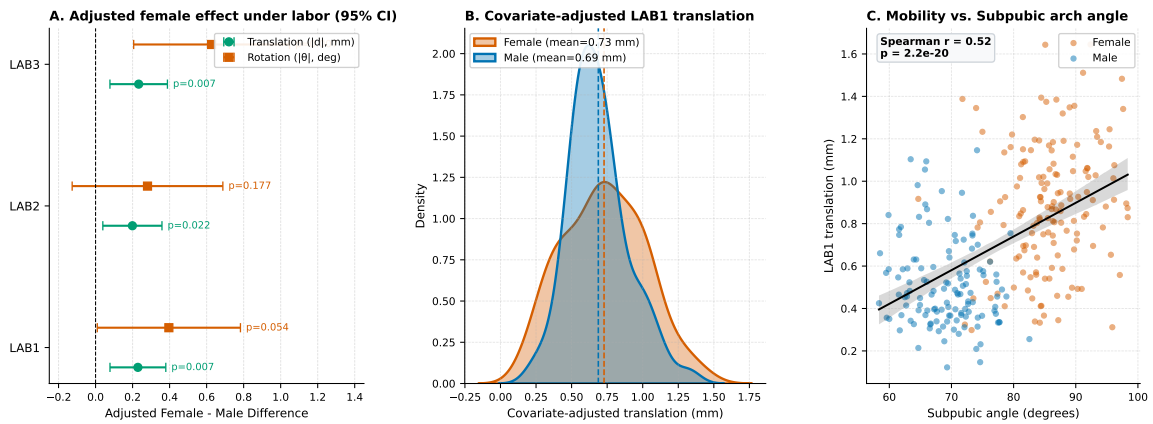


Figure 4: Sex-related differences under parturition loading and morphometric associations. (A) Forest plot of adjusted female-minus-male differences (95% CI) for translation magnitude (green circles) and rotation magnitude (orange squares) across LAB1–LAB3 from multivariable robust OLS models. (B) Kernel density distributions of covariate-adjusted translation in LAB1 for females (vermilion) and males (blue). (C) Morphometric association scatterplot of LAB1 translation magnitude versus subpubic arch angle across the cohort ($r = 0.52$, $p = 2.2 \times 10^{-20}$).

3.5 Variance channels: structural geometry versus material heterogeneity (H4)

Decomposition of between-subject variance across the three matched simulation variants revealed that macroscopic skeletal geometry accounts for the predominant share of population dispersion

(Fig. 5; Supplementary Table S2). The shape-only model reproduced 98.7% to 102.1% (overall median 100.46%) of full-model variance across all load cases and kinematic degrees of freedom. Values slightly exceeding 100% reflect sample-variance ratios derived from separate simulation datasets, signifying near-complete variance retention by skeletal geometry.

Paired subject-level diagnostics between the Full and Shape-Only models confirmed close individual correspondence across all load regimes. For resultant translation magnitude, the coefficient of determination against the 1:1 identity line was $R^2 \geq 0.994$, with Pearson correlation $r \geq 0.997$, Spearman rank correlation $r_s \geq 0.996$, RMSE ≤ 0.031 mm, and MAE ≤ 0.021 mm. For resultant rotation, concordance was similarly high ($R^2 \geq 0.999$, $r \geq 0.999$, RMSE $\leq 0.023^\circ$, MAE $\leq 0.015^\circ$).

In contrast, the material-only model (isolating CT-derived bone mineral density and elastic modulus heterogeneity on the unmapped reference geometry) accounted for a small fraction of population variance, ranging from 0.001% to 0.317% (overall median 0.033%). For resultant rotation and translation, material-only variance fractions remained below 0.053% and 0.295%, respectively. Under parturition-motivated configurations, material-only variance did not exceed 0.067% for translation or 0.030% for rotation. These results indicate that inter-individual differences in bone mineral density have a minor influence on macroscopic joint micromotion under standardized loading compared to skeletal architecture.

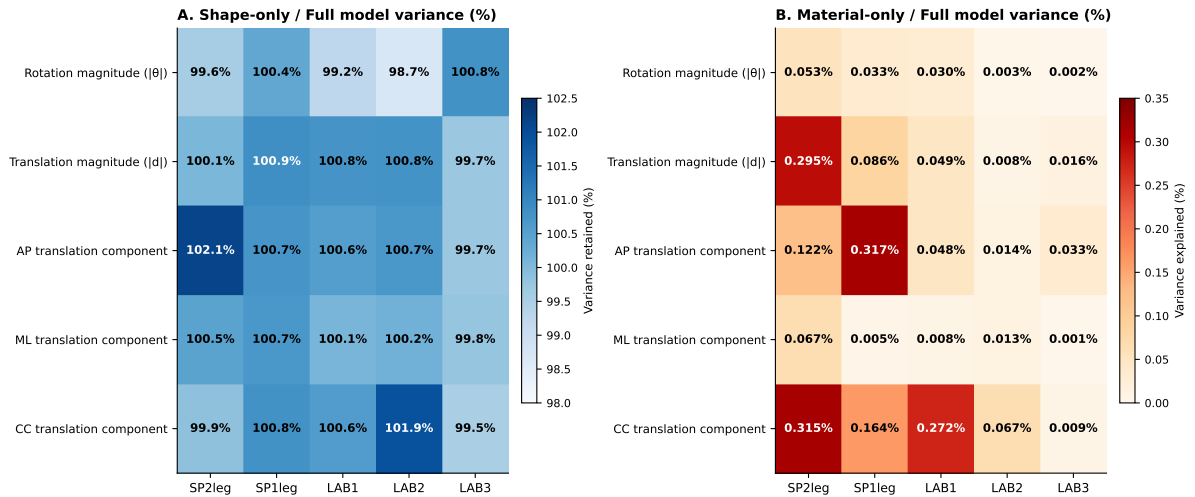


Figure 5: Variance-channel decomposition: shape versus bone material heterogeneity. Dual annotated percentage heatmaps displaying: (A) Proportion of full-model population variance retained by the shape-only model ($V_{\text{shape}}/V_{\text{full}} \times 100\%$). (B) Proportion of full-model variance explained by the material-only model ($V_{\text{material}}/V_{\text{full}} \times 100\%$). Cell annotations indicate exact percentages across five load cases and six kinematic metrics.

3.6 Secondary allometric scaling with pelvic size

Log-log power-law regressions of SIJ micromotion against total pelvic bone volume revealed load-dependent scaling (Table 4B; Fig. 6). Statistically significant negative allometric scaling was observed during single-leg stance (SP1leg): larger pelvic volume was associated with reduced micromotion in both sexes, with scaling exponents of $\beta_{\text{male}} = -0.482$ (95% CI -0.870 to -0.094 , raw $p = 0.0148$, FDR $q = 0.0742$) and $\beta_{\text{female}} = -0.428$ (95% CI -0.762 to -0.093 , raw $p = 0.0122$, FDR $q = 0.0405$) for rotation ($R^2 = 0.38$), and $\beta_{\text{male}} = -0.569$ (95% CI -0.869 to -0.269 , raw $p < 0.001$, FDR $q = 0.0020$) and $\beta_{\text{female}} = -0.432$ (95% CI -0.669 to -0.194 , raw $p < 0.001$, FDR $q = 0.0037$) for translation ($R^2 = 0.50$).

Under two-leg stance and parturition configurations, volume scaling exponents were generally attenuated, with 95% confidence intervals crossing zero in most cases. A notable exception was female translation in LAB2, which retained a statistically significant negative scaling exponent ($\beta_{\text{female}} = -0.614$, 95% CI -1.061 to -0.166 , raw $p = 0.0072$, BH-FDR $q = 0.0360$), whereas male scaling in LAB2 was not significant ($\beta_{\text{male}} = -0.367$, 95% CI -0.904 to 0.171 , $p = 0.1809$, $q = 0.3654$). Crucially, sex-by-size interaction terms were not statistically significant across any load case or outcome after BH-FDR correction (all corrected $q \geq 0.958$), indicating that male and female pelves follow consistent allometric scaling trajectories.

In pooled cluster-robust models across all 1,390 observations, translation magnitude exhibited an overall negative volume association ($\beta = -0.092$ per 1 SD increase, 95% CI -0.143 to -0.041 , BH-FDR $q < 0.001$; Table 4A), modulated by significant load $\times$ size interaction terms.

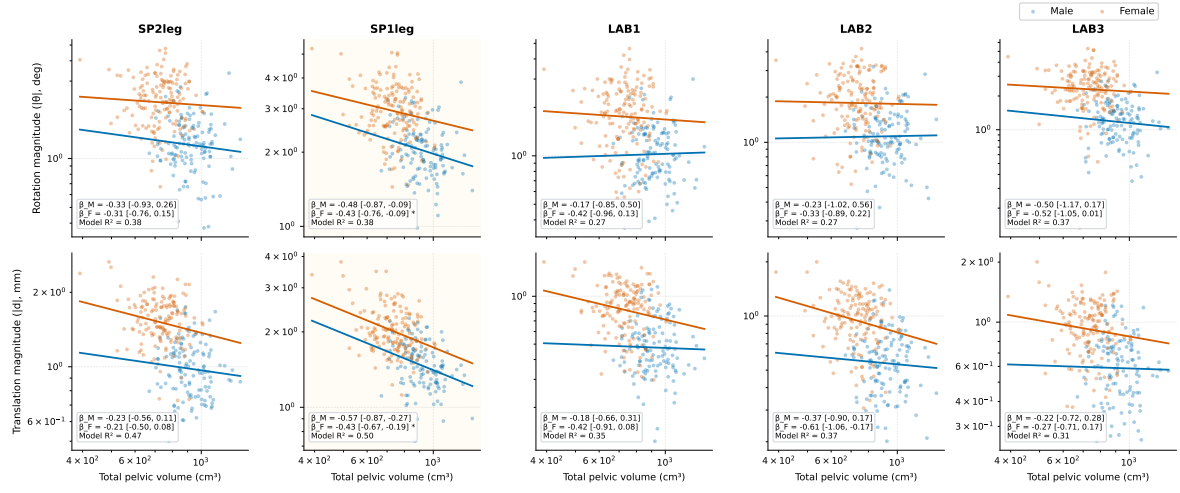


Figure 6: **Allometric log–log scaling of SIJ micromotion with total pelvic volume.** 2×5 regression grid for rotation magnitude (top) and translation magnitude (bottom) across SP2leg, SP1leg, and LAB1–LAB3. Points show male (blue) and female (vermillion) subjects; solid lines depict sex-stratified power-law fits. The shaded panel highlights single-leg stance (SP1leg), which displays consistent negative allometric scaling in both sexes.

Table 4: Pooled cluster-robust regression models and load-wise allometric scaling exponents. Panel A: Pooled models across all 1,390 observations (clustered by subject). Panel B: Load-specific log-log power-law scaling exponents (β and 95% CI).

Panel A

Outcome	Predictor	β [95% CI]	Std. β	Partial R^2	q_{FDR}
Rotation magnitude ($ \theta $, deg)	SP1leg (vs SP2leg)	0.723 [0.672, 0.773]	0.781	0.362	< 0.001
	LAB1 (vs SP2leg)	-0.326 [-0.359, -0.292]	-0.352	0.209	< 0.001
	LAB2 (vs SP2leg)	-0.261 [-0.323, -0.198]	-0.282	0.046	< 0.001
	LAB3 (vs SP2leg)	0.060 [0.033, 0.087]	0.065	0.013	< 0.001
	Female sex	0.450 [0.103, 0.797]	0.487	0.005	0.020
	Age (z)	0.044 [-0.053, 0.141]	0.048	0.001	0.398
	Pelvic volume (z)	-0.126 [-0.285, 0.034]	-0.136	0.002	0.182
	SP1leg $\times$ Volume (z)	-0.032 [-0.079, 0.015]	-0.035	0.001	0.223
	LAB1 $\times$ Volume (z)	0.096 [0.062, 0.130]	0.104	0.022	< 0.001
	LAB2 $\times$ Volume (z)	0.105 [0.033, 0.177]	0.114	0.006	0.009
	LAB3 $\times$ Volume (z)	-0.065 [-0.093, -0.038]	-0.071	0.016	< 0.001
	Inlet AP diameter (z)	0.130 [0.028, 0.232]	0.141	0.005	0.020
	Biischadic width (z)	0.082 [-0.035, 0.200]	0.089	0.001	0.223
	Subpubic angle (z)	0.079 [-0.075, 0.232]	0.085	0.001	0.366
	Female $\times$ Volume (z)	-0.079 [-0.280, 0.123]	-0.085	0.000	0.443
Translation magnitude ($ \mathbf{d} $, mm)	SP1leg (vs SP2leg)	0.499 [0.465, 0.533]	0.884	0.375	< 0.001
	LAB1 (vs SP2leg)	-0.598 [-0.627, -0.569]	-1.060	0.542	< 0.001
	LAB2 (vs SP2leg)	-0.498 [-0.533, -0.462]	-0.882	0.353	< 0.001
	LAB3 (vs SP2leg)	-0.494 [-0.513, -0.475]	-0.875	0.659	< 0.001
	Female sex	0.254 [0.121, 0.387]	0.450	0.010	< 0.001
	Age (z)	0.028 [-0.008, 0.063]	0.049	0.002	0.134
	Pelvic volume (z)	-0.092 [-0.143, -0.041]	-0.163	0.009	< 0.001
	SP1leg $\times$ Volume (z)	-0.073 [-0.106, -0.040]	-0.129	0.013	< 0.001
	LAB1 $\times$ Volume (z)	0.073 [0.043, 0.103]	0.130	0.016	< 0.001
	LAB2 $\times$ Volume (z)	0.048 [0.008, 0.088]	0.085	0.004	0.025
	LAB3 $\times$ Volume (z)	0.085 [0.068, 0.102]	0.150	0.063	< 0.001
	Inlet AP diameter (z)	-0.062 [-0.095, -0.028]	-0.109	0.009	< 0.001
	Biischadic width (z)	0.043 [-0.002, 0.089]	0.076	0.002	0.079
	Subpubic angle (z)	0.051 [-0.012, 0.114]	0.091	0.002	0.129
	Female $\times$ Volume (z)	-0.039 [-0.111, 0.034]	-0.069	0.001	0.293

Panel B

Load case	Outcome	Male exponent [95% CI]	Female exponent [95% CI]	Interaction β	q_{FDR}	Adj. R^2
SP2leg	Rotation ($ \theta $)	-0.334 [-0.931, 0.263]	-0.306 [-0.762, 0.151]	0.028	0.958	0.383
	Translation ($ \mathbf{d} $)	-0.227 [-0.562, 0.107]	-0.211 [-0.505, 0.084]	0.017	0.958	0.468
SP1leg	Rotation ($ \theta $)	-0.482 [-0.870, -0.094]	-0.428 [-0.762, -0.093]	0.055	0.958	0.382
	Translation ($ \mathbf{d} $)	-0.569 [-0.869, -0.269]	-0.432 [-0.669, -0.194]	0.137	0.958	0.500
LAB1	Rotation ($ \theta $)	-0.172 [-0.846, 0.502]	-0.416 [-0.959, 0.127]	-0.244	0.958	0.267
	Translation ($ \mathbf{d} $)	-0.177 [-0.665, 0.311]	-0.419 [-0.914, 0.077]	-0.242	0.958	0.354
LAB2	Rotation ($ \theta $)	-0.228 [-1.020, 0.565]	-0.334 [-0.891, 0.223]	-0.106	0.958	0.269
	Translation ($ \mathbf{d} $)	-0.367 [-0.904, 0.171]	-0.614 [-1.061, -0.166]	-0.247	0.958	0.375
LAB3	Rotation ($ \theta $)	-0.498 [-1.169, 0.174]	-0.518 [-1.048, 0.011]	-0.021	0.958	0.367
	Translation ($ \mathbf{d} $)	-0.220 [-0.720, 0.280]	-0.271 [-0.708, 0.166]	-0.051	0.958	0.310

3.7 Summary of hypothesis testing

In summary:

- **H1 (Load-Dependence and Stance Asymmetry): Supported.** Locomotor loading engaged self-bracing nutation ($\theta_{\text{nut}} \geq 1.06^\circ$), with SP1leg driving a substantial increase in bilateral translation asymmetry (2.01 mm vs. 0.21 mm) and frontal tilt. Parturition proxies reduced sagittal nutation (0.41° – 0.43°) and re-routed displacement into transverse expansion.
- **H2 (Directional Kinematic Contrasts): Supported.** Parturition loading reduced vertical craniocaudal shear ($\Delta\text{CC} \leq -0.85$ mm) while selectively augmenting transverse distraction ($\Delta\text{ML} \geq +0.09$ mm) and sagittal gliding.
- **H3 (Sex-Related Differences and Morphometric Associations): Supported.** Female pelves exhibited greater translational compliance across labor configurations (+0.19 to +0.23 mm in fully adjusted models, $p \leq 0.020$). This difference attenuated upon adjusting for pelvic size and architecture, and correlated positively with subpubic arch angle ($r = 0.52, p < 10^{-19}$).
- **H4 (Relative Contributions of Shape versus Material Heterogeneity): Supported.** Macroscopic skeletal geometry accounted for 98.7% to 102.1% (median 100.5%) of population variance, whereas bone mineral density heterogeneity explained less than 0.32% (median 0.033%). Allometric volume scaling was load-dependent, manifesting most clearly during single-leg stance.

4 Discussion

By analyzing a standardized finite-element repository of 278 human pelves parameterized via an implicit continuous mapping, this study provides a population-scale quantitative evaluation of sacroiliac joint (SIJ) micromotion across locomotor and parturition-motivated mechanical regimes. The findings provide insight into how the passive architecture of the human pelvic ring accommodates the contrasting mechanical demands of bipedal weight-bearing stability and birth canal clearance (Washburn, 1960; Ruff, 2017; Pavličev et al., 2020). Four central observations emerge: (1) SIJ kinematics display marked load-dependence, engaging self-bracing nutation under bilateral stance and pronounced asymmetric shear under single-leg stance, while

parturition-motivated configurations reduce nutation; (2) obstetric mechanical proxies shift displacement from vertical craniocaudal shear toward transverse outward distraction; (3) female pelves exhibit greater translational compliance under simulated labor, an effect that is attenuated when adjusting for pelvic volume and transverse arch architecture; and (4) between-subject kinematic dispersion is governed predominantly by macro-architectural skeletal geometry, with bone mineral density heterogeneity playing a minor role ($< 0.32\%$).

Functional morphology: form and force closure under locomotor demand

Our findings provide quantitative support for the biomechanical framework of *form and force closure* described by Vleeming et al. (2012) and Snijders et al. (1993). Under upright bilateral stance (SP2leg), vertical ground reaction forces transmitted through both acetabula against the fixed sacrum engage physiological nutation (median $\theta_{\text{nut}} = 1.062^\circ$), rotating the sacral base anteriorly and inferiorly relative to the ilia. This sagittal tilt increases tension across the posterior interosseous and sacrotuberous ligamentous apparatus, drawing the articular surfaces into closer apposition and wedging the sacrum within the pelvic ring (Snijders et al., 1993; Hammer et al., 2013; Kiapour et al., 2019).

During single-leg stance (SP1leg), this self-bracing mechanism is subjected to asymmetric loading. Ground reaction forces transmitted through a single femoral head create an uncoupled reaction couple, driving resultant translation to 1.755 mm and generating a substantial increase in bilateral translation asymmetry (median 2.013 mm vs. 0.215 mm in bilateral stance; Fig. 2D). This asymmetry is accompanied by frontal pelvic tilt (abduction $\theta_y = 1.582^\circ$) and vertical shear ($\Delta\text{CC} = +0.394$ mm), as the unsupported contralateral hemi-pelvis drops while the loaded hemi-pelvis shears superiorly. This marked mechanical disparity is consistent with clinical observations that single-limb weight-bearing tasks (such as single-leg stance, stair ascent, and running) frequently provoke symptoms in patients with sacroiliac joint dysfunction and pelvic girdle instability (Goode et al., 2008; Hammer et al., 2019; Keizer et al., 2019).

Directional kinematic contrasts and mechanical proxies of parturition

A key observation of this study is that evaluating SIJ response solely by scalar displacement magnitude provides an incomplete picture of joint mechanics. While parturition-motivated load configurations (LAB1–LAB3) produce lower scalar translation magnitudes (0.66–0.79 mm) than bipedal standing (1.30 mm), decomposition into local anatomical components reveals directional

re-routing (Fig. 3). In contrast to standing, internal ring compression reduces sagittal nutation (θ_{nut} decreases to 0.41° – 0.43°) and diminishes craniocaudal shear ($\Delta\text{CC} = -0.85$ to -0.90 mm). Instead, joint displacements are directed primarily into transverse outward distraction ($\Delta\text{ML} = +0.09$ to $+0.15$ mm) and sagittal gliding ($\Delta\text{AP} = +0.16$ mm).

This directional shift provides a mechanical perspective relevant to evolutionary discussions of the obstetrical dilemma (Washburn, 1960; Ruff, 2017; Pavličev et al., 2020). To accommodate the passage of a neonate, the female pelvic ring must allow transient transverse and anteroposterior cavity dilation (Mora et al., 2022; Fuchs et al., 2013; Roque de Almeida et al., 2024). If parturition forces primarily engaged vertical shear or excessive counternutation, articular cartilages and dorsal ligaments would experience severe localized strain. By channeling internal expansion forces into transverse compliance (ΔML) while reducing vertical shear, the passive architecture of the pelvic ring facilitates outward dilation of the cavity while maintaining joint congruency. We emphasize, however, that the static forces applied here serve as standardized mechanical proxies and do not capture the dynamic, contact-driven progression of the fetal head through the birth canal.

Sex-related differences and morphometric associations

Multivariable regression models demonstrated a statistically significant female translational surplus across all labor configurations ($+0.19$ to $+0.23$ mm in fully adjusted models, $p \leq 0.020$; Fig. 4). Importantly, our nested regression analysis showed that this surplus attenuates by approximately 38% upon adjusting for total pelvic volume and the morphometric triad (AP diameter, biischadic width, and subpubic arch angle). This indicates that a substantial component of the observed sex-related difference is explained by sexual dimorphism in pelvic size and anterior arch architecture.

Furthermore, compliance in LAB1 was positively associated with the subpubic arch angle (Spearman $r = 0.52$, $p = 2.2 \times 10^{-20}$; Fig. 4C). Anatomically, the subpubic arch forms the anterior arch of the lower pelvic ring (Ruff, 2017; Betti and Manica, 2020; Weiss, 2020). A broader subpubic angle (85.8° in females vs. 69.1° in males; Table 1) increases the transverse span relative to the pubic symphysis, decreasing the hoop stiffness of the anterior ring. Consequently, internal transverse expansion forces produce greater bilateral displacement at the dorsal SIJs in female pelvis. This architectural relationship provides a structural hypothesis for future investigation of peripartum pelvic girdle pain (PGP), where broader pelvic dimensions may

coincide with greater passive compliance under mechanical loading (Hammer et al., 2019; Keizer et al., 2019).

Skeletal geometry dominates bone material heterogeneity

A central methodological finding of this work is the variance-channel comparison across matched simulation cohorts (Fig. 5). Between-subject dispersion in SIJ micromotion was governed predominantly by macroscopic skeletal geometry: the shape-only model accounted for 98.7% to 102.1% (median 100.5%) of full-model population variance, with paired subject-level diagnostics showing near-perfect agreement ($R^2 \geq 0.994$, Pearson $r \geq 0.997$, RMSE ≤ 0.031 mm). In contrast, the material-only model accounted for less than 0.32% (median 0.033%) of population variance.

This disparity has a solid basis in structural mechanics. The human pelvis functions as an indeterminate closed arch ring. The flexural and torsional compliances of an arch ring scale with the cube of cortical thickness and the geometry of the cross-section ($I \propto b \cdot h^3$), whereas material modulus E enters linearly in the denominator. Across the adult population, inter-individual variations in pelvic dimensions, cortical curvatures, and articular orientations vary sufficiently to dominate load routing, rendering the influence of local bone mineral density heterogeneity secondary under standardized external loading.

Importantly, this comparison specifically isolates *bone mineral density (modulus) heterogeneity* derived from clinical CT scans, while keeping cartilage and ligament stiffness nominal. These results suggest that for macroscopic joint kinematic predictions, accurate subject-specific anatomical segmentation is paramount, whereas simplified bone density distributions introduce minimal bias into kinematic endpoints.

Secondary allometric size scaling

Secondary log-log regressions indicated that allometric size scaling is load-dependent (Table 4; Fig. 6). Statistically significant negative allometric volume scaling was observed during single-leg stance (SP1leg: $\beta_{\text{male}} = -0.57$, $\beta_{\text{female}} = -0.43$, $q \leq 0.004$ for translation), indicating that larger pelvic volume provides greater structural stiffness against asymmetric vertical shear. Under two-leg stance and parturition configurations, volume scaling exponents were generally attenuated, although female translation in LAB2 retained a statistically significant negative slope ($\beta = -0.614$, $q = 0.0360$).

Importantly, sex-by-size interaction terms were not statistically significant across any loading regime ($q \geq 0.958$), indicating that male and female pelvises follow consistent scaling trajectories. Sex-related differences in SIJ compliance thus reflect an intercept shift related to dimorphic arch geometry rather than distinct scaling exponents.

Biomechanical and clinical considerations

These findings provide mechanical context for orthopedic and pelvic health considerations:

1. **Sacroiliac Joint Arthrodesis:** Minimally invasive SIJ fusion is utilized for chronic, recalcitrant joint pain. Our data indicate that the greatest kinematic demand occurs during unilateral stance, marked by a substantial increase in bilateral shear and frontal tilt. These findings suggest that surgical fixation strategies should prioritize resistance to asymmetric craniocaudal shear and torsional tilt rather than symmetric axial compression (Kiapour et al., 2019; Anderson et al., 2020).

2. **Pelvic Girdle Compliance:** The positive correlation between subpubic arch angle and transverse compliance ($r = 0.52$) highlights the role of anterior ring geometry in posterior joint mechanics. While external pelvic compression belts are clinically applied to enhance force closure, further biomechanical studies are needed to determine whether individuals with broader arch dimensions experience differential benefit from external bracing (Vleeming et al., 2012; Hammer et al., 2013).

Strengths and limitations

The strengths of this study include the evaluation of a standardized 278-subject finite-element cohort across an adult lifespan, complete morphometric profiling, matched simulation variants isolating shape from bone material heterogeneity, and robust multivariable modeling with multiplicity control.

Several limitations must be acknowledged. First, simulations were formulated under static small-strain continuum elastostatics with linear-elastic bone, cartilage, and symphysis representations; dynamic inertial effects during gait and viscoelastic soft-tissue relaxation were not modeled. Second, the parturition load cases represent standardized static mechanical proxies rather than dynamic simulations of fetal head engagement and soft-tissue contact (Chen and Grimm, 2021; Roque de Almeida et al., 2024; Mora et al., 2022). Third, the models did not in-

corporate pregnancy-induced hormonal laxity (e.g., relaxin-mediated ligament softening), which would be expected to augment compliance in vivo. Consequently, the observed female surplus reflects baseline skeletal and passive ligamentous architecture. Incorporating subject-specific soft-tissue properties and dynamic contact formulations represents an important area for future research.

5 Conclusion

In a standardized cohort of 278 human pelves evaluated under finite-element loading, sacroiliac joint micromotion revealed four fundamental biomechanical principles:

1. **Load-Dependence and Stance Asymmetry:** Erect bipedal standing reinforces form and force closure through physiological nutation ($\theta_{\text{nut}} = 1.062^\circ$), whereas single-leg stance imposes the highest kinematic demand among the evaluated regimes, driving a marked increase in bilateral translation asymmetry (2.013 mm vs. 0.215 mm) and frontal tilt.
2. **Directional Kinematic Contrasts:** Parturition-motivated forces suppress sagittal nutation and vertical shear, shifting joint displacement toward transverse outward distraction ($\Delta\text{ML} \geq +0.09$ mm) and sagittal gliding, illustrating passive compliance pathways that facilitate cavity expansion while maintaining articular congruency.
3. **Sex-Related Differences and Architectural Associations:** Female pelves exhibit significantly greater translational compliance across labor proxies (+0.19 to +0.23 mm in fully adjusted models, $p \leq 0.020$). This difference attenuates upon adjusting for pelvic volume and dimensions, and correlates positively with subpubic arch angle ($r = 0.52, p < 10^{-19}$) due to the lower hoop stiffness of broader anterior arch morphology.
4. **Dominance of Skeletal Geometry over Bone Mineralization Heterogeneity:** Macro-architectural pelvic anatomy accounts for 98.7% to 102.1% (median 100.5%) of between-subject kinematic variance ($R^2 \geq 0.994$), whereas bone mineral density heterogeneity explains less than 0.32% (median 0.033%), establishing that 3D skeletal geometry, rather than bone mineralization, governs joint-level kinematic boundaries.

Together, these findings demonstrate that sacroiliac joint kinematics are governed by the interplay of load configuration and skeletal architecture, clarifying the mechanical compromise

between bipedal stability and pelvic compliance, and providing quantitative anatomical benchmarks for orthopedic reconstruction and pelvic biomechanics.

A Mathematical Formulation of Sacroiliac Joint Micromotion Analysis

This appendix provides a self-contained, mathematically rigorous specification of the computational pipeline used to extract three-dimensional relative sacroiliac joint (SIJ) kinematics, boundary conditions, and morphometric dimensions from the finite-element (FE) cohort simulations.

A.1 Continuum finite-element formulation and boundary value problem

The mechanical equilibrium of the human pelvic ring is governed by the weak form of small-strain linear elasticity:

$$\int_{\Omega} \boldsymbol{\sigma}(\mathbf{u}) : \boldsymbol{\varepsilon}(\mathbf{v}) d\Omega + \int_{\Gamma_{S1}} \gamma (\mathbf{u} \cdot \mathbf{v}) ds = \int_{\Gamma_{load}} \mathbf{t}_{ext} \cdot \mathbf{v} ds, \quad \forall \mathbf{v} \in \mathcal{V}, \quad (8)$$

where $\Omega \subset \mathbb{R}^3$ denotes the pelvic continuum domain, $\mathbf{u}(\mathbf{x})$ is the displacement field, $\mathbf{v}$ is an arbitrary virtual displacement test function in the Sobolev space $\mathcal{V} = [H^1(\Omega)]^3$, $\boldsymbol{\varepsilon}(\mathbf{u}) = \frac{1}{2}(\nabla \mathbf{u} + (\nabla \mathbf{u})^T)$ is the infinitesimal strain tensor, and $\boldsymbol{\sigma} = \mathbf{C} : \boldsymbol{\varepsilon}$ is the Cauchy stress tensor.

Fixed anatomical boundary conditions are enforced on the superior S1 sacral facet (Γ_{S1}) via a high-stiffness Dirichlet penalty parameter:

$$\gamma = 10^6 \text{ N/mm}, \quad (9)$$

which effectively enforces $\mathbf{u}|_{\Gamma_{S1}} \approx \mathbf{0}$ while preserving variational symmetry. External static load cases apply surface tractions $\mathbf{t}_{ext}$ integrated over specified anatomical surface facets Γ_{load} such that $\int_{\Gamma_{load}} \mathbf{t}_{ext} ds = \mathbf{F}_{ext}$:

1. **Bilateral Stance (SP2leg):** Upward ground reaction forces applied to bilateral acetabular notches:

$$\mathbf{F}_{AC,L} = [0, 0, +400]^T \text{ N}, \quad \mathbf{F}_{AC,R} = [0, 0, +400]^T \text{ N} \quad (\Sigma F_{CC} = +800 \text{ N}). \quad (10)$$

2. **Unilateral Stance (SP1leg):** Upward ground reaction force applied to the right acetabular notch:

$$\mathbf{F}_{AC,R} = [0, 0, +800]^T \text{ N}, \quad \mathbf{F}_{AC,L} = \mathbf{0}. \quad (11)$$

3. **Internal Ring Compression/Shear Proxy (LAB1):** Opposing mediolateral force pair applied to the inner pectineal pelvic ring surfaces:

$$\mathbf{F}_{ring,L} = [+400, 0, 0]^T \text{ N}, \quad \mathbf{F}_{ring,R} = [-400, 0, 0]^T \text{ N}. \quad (12)$$

4. **Ischial Distraction Proxy (LAB2):** Opposing mediolateral outward force pair applied to bilateral ischial tuberosities:

$$\mathbf{F}_{isch,L} = [+400, 0, 0]^T \text{ N}, \quad \mathbf{F}_{isch,R} = [-400, 0, 0]^T \text{ N}. \quad (13)$$

5. **Outlet Sagittal Distraction Proxy (LAB3):** Opposing anteroposterior force pair applied to the anterior pubic insertion and posterior sacrococcygeal junction (SCJ):

$$\mathbf{F}_{pubis} = [0, -400, 0]^T \text{ N}, \quad \mathbf{F}_{SCJ} = [0, +400, 0]^T \text{ N}. \quad (14)$$

Constitutive materials comprise CT-derived heterogeneous Young's modulus $E(\mathbf{x})$ for cortical/trabecular bone ($\nu = 0.3$), isotropic articular cartilage ($E = 15 \text{ MPa}$, $\nu = 0.45$), pubic symphyseal disc ($E = 5 \text{ MPa}$, $\nu = 0.45$), and 1D tension-only ligament bundles with anatomically assigned stiffnesses and initial pretension.

A.2 Construction of the anatomical pelvic coordinate frame

To ensure objective, standardized kinematic reporting invariant to global spatial placement, an anatomical pelvic coordinate frame $\mathbf{P} \in \mathbb{R}^{3 \times 3}$ is constructed from the reference pelvic geometry $\mathcal{P}^0 = \{\mathbf{p}_i^0\}_{i=1}^N$:

1. **Midsagittal Symmetry Plane:** The global mediolateral unit normal $\hat{\mathbf{x}}_{ML}$ and pelvic origin $\mathbf{o} \in \mathbb{R}^3$ are determined by identifying the optimal mirror symmetry plane minimizing bilateral reflection distance across the pelvic point cloud.

2. **In-Plane Projection and Craniocaudal Axis:** Pelvic coordinates are projected onto

the midsagittal symmetry plane:

$$\mathbf{p}_i^{\text{proj}} = (\mathbf{p}_i^0 - \mathbf{o}) - [(\mathbf{p}_i^0 - \mathbf{o}) \cdot \hat{\mathbf{x}}_{\text{ML}}] \hat{\mathbf{x}}_{\text{ML}}. \quad (15)$$

Singular value decomposition (SVD) on the centered in-plane coordinates $\mathbf{X} = \mathbf{p}_i^{\text{proj}} - \bar{\mathbf{p}}^{\text{proj}} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T$ extracts the dominant longitudinal principal direction $\mathbf{v}_1$. Orthogonalizing against $\hat{\mathbf{x}}_{\text{ML}}$ establishes the craniocaudal unit vector $\hat{\mathbf{z}}_{\text{CC}}$:

$$\hat{\mathbf{z}}_{\text{CC}} = \frac{\mathbf{v}_1 - (\mathbf{v}_1 \cdot \hat{\mathbf{x}}_{\text{ML}}) \hat{\mathbf{x}}_{\text{ML}}}{\|\mathbf{v}_1 - (\mathbf{v}_1 \cdot \hat{\mathbf{x}}_{\text{ML}}) \hat{\mathbf{x}}_{\text{ML}}\|}. \quad (16)$$

3. Anteroposterior Axis and Right-Handed System: The anteroposterior unit vector $\hat{\mathbf{y}}_{\text{AP}}$ is established via the vector cross product:

$$\hat{\mathbf{y}}_{\text{AP}} = \hat{\mathbf{z}}_{\text{CC}} \times \hat{\mathbf{x}}_{\text{ML}}, \quad \text{with } \hat{\mathbf{z}}_{\text{CC}} \leftarrow \hat{\mathbf{x}}_{\text{ML}} \times \hat{\mathbf{y}}_{\text{AP}}. \quad (17)$$

4. Directional Consistency: The sign of $\hat{\mathbf{x}}_{\text{ML}}$ is aligned such that it points from anatomical left to right: if $(\bar{\mathbf{p}}_{\text{left}} - \mathbf{o}) \cdot \hat{\mathbf{x}}_{\text{ML}} > (\bar{\mathbf{p}}_{\text{right}} - \mathbf{o}) \cdot \hat{\mathbf{x}}_{\text{ML}}$, then $\hat{\mathbf{x}}_{\text{ML}} \leftarrow -\hat{\mathbf{x}}_{\text{ML}}$ and $\hat{\mathbf{y}}_{\text{AP}} \leftarrow -\hat{\mathbf{y}}_{\text{AP}}$.

The resulting orthonormal transformation matrix is:

$$\mathbf{P} = \begin{bmatrix} \hat{\mathbf{x}}_{\text{ML}} & \hat{\mathbf{y}}_{\text{AP}} & \hat{\mathbf{z}}_{\text{CC}} \end{bmatrix}, \quad \mathbf{P}^T \mathbf{P} = \mathbf{I}_{3 \times 3}. \quad (18)$$

A.3 Global affine sacrum drift elimination

To isolate intra-pelvic joint mobility from any slight residual rigid-body motion or global affine deformation of the sacral base, an affine transformation $(\mathbf{A}_S, \mathbf{b}_S)$ is fitted between reference sacral coordinates $\mathbf{P}_S^0 \in \mathbb{R}^{N_S \times 3}$ and deformed sacral coordinates $\mathbf{P}_S^1 \in \mathbb{R}^{N_S \times 3}$:

$$\min_{\mathbf{A}_S \in \mathbb{R}^{3 \times 3}, \mathbf{b}_S \in \mathbb{R}^3} \sum_{i=1}^{N_S} \|\mathbf{p}_{S,i}^1 - (\mathbf{A}_S \mathbf{p}_{S,i}^0 + \mathbf{b}_S)\|^2. \quad (19)$$

The optimal affine map is obtained via regularized least squares. The inverse affine transformation is then applied to all deformed bones (sacrum, left ilium, right ilium):

$$\mathbf{x}' = \mathbf{A}_S^{-1}(\mathbf{x}^1 - \mathbf{b}_S), \quad (20)$$

602 anchoring the sacrum to its un-deformed reference baseline.

603 A.4 Articular region of interest (ROI) identification

604 For each joint side $k \in \{\text{left}, \text{right}\}$, opposing articulating bone surfaces are isolated using a
 605 proximity threshold $g = 5.0$ mm:

$$606 \quad \mathcal{R}_S^{(k)} = \left\{ \mathbf{p} \in \mathcal{S}_{\text{sac}}^{(k)} \mid \min_{\mathbf{q} \in \mathcal{S}_{\text{ili}}^{(k)}} \|\mathbf{p} - \mathbf{q}\| \leq g \right\}, \quad (21)$$

$$607 \quad \mathcal{R}_I^{(k)} = \left\{ \mathbf{q} \in \mathcal{S}_{\text{ili}}^{(k)} \mid \min_{\mathbf{p} \in \mathcal{S}_{\text{sac}}^{(k)}} \|\mathbf{q} - \mathbf{p}\| \leq g \right\}. \quad (22)$$

608 If an extracted ROI contains fewer than $N_{\min} = 80$ nodes, the $N_{\min}$ nearest surface nodes are
 609 selected to ensure robust registration support.

610 A.5 Robust trimmed Kabsch rigid registration

611 For both articulating bones (sacrum S and ilium I), rigid transformations $(\mathbf{R}_B, \mathbf{t}_B)$ mapping
 612 reference ROI points $\mathbf{P}_B^0$ to transformed deformed ROI points $\mathbf{Q}'_B$ are determined using the
 613 Kabsch algorithm with iterative outlier trimming (`trim_frac` = 0.10, 2 iterations).

614 Given centered coordinates $\tilde{\mathbf{P}} = \mathbf{P}_B^0 - \bar{\mathbf{p}}_B^0$ and $\tilde{\mathbf{Q}} = \mathbf{Q}'_B - \bar{\mathbf{q}}'_B$, the cross-covariance matrix is:

$$615 \quad \mathbf{C} = \tilde{\mathbf{P}}^T \tilde{\mathbf{Q}} \in \mathbb{R}^{3 \times 3}. \quad (23)$$

616 Computing the singular value decomposition $\mathbf{C} = \mathbf{U} \mathbf{\Sigma} \mathbf{V}^T$, the optimal rotation matrix is:

$$617 \quad \mathbf{R}_B = \mathbf{V} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & d \end{bmatrix} \mathbf{U}^T, \quad \text{where } d = \det(\mathbf{V} \mathbf{U}^T) \in \{+1, -1\}, \quad (24)$$

618 which strictly enforces $\det(\mathbf{R}_B) = +1$ and prevents improper reflection. The translation vector
 619 is:

$$620 \quad \mathbf{t}_B = \bar{\mathbf{q}}'_B - \mathbf{R}_B \bar{\mathbf{p}}_B^0. \quad (25)$$

621 Residual spatial deviations $r_i = \|\mathbf{R}_B \mathbf{p}_{B,i}^0 + \mathbf{t}_B - \mathbf{q}'_{B,i}\|$ are evaluated across all points; the
 622 top 10% largest residuals are trimmed, and $(\mathbf{R}_B, \mathbf{t}_B)$ is recomputed on the remaining inliers.

623 This prevents localized superficial element distortion from biasing macroscopic rigid-body joint

estimates.

A.6 Relative kinematics and Cardan angle decomposition

The relative rigid transformation of the ilium with respect to the sacrum is given by:

$$\mathbf{R}_{\text{rel}} = \mathbf{R}_S^T \mathbf{R}_I, \quad \mathbf{t}_{\text{rel}} = \mathbf{R}_S^T (\mathbf{t}_I - \mathbf{t}_S). \quad (26)$$

Projecting the relative rotation into the anatomical pelvic coordinate system yields:

$$\mathbf{R}_{\text{loc}} = \mathbf{P}^T \mathbf{R}_{\text{rel}} \mathbf{P} = \begin{bmatrix} R_{00} & R_{01} & R_{02} \\ R_{10} & R_{11} & R_{12} \\ R_{20} & R_{21} & R_{22} \end{bmatrix}. \quad (27)$$

Factoring $\mathbf{R}_{\text{loc}}$ into an intrinsic Cardan XYZ (Tait-Bryan) sequence:

$$\mathbf{R}_{\text{loc}} = \mathbf{R}_x(\alpha_x) \mathbf{R}_y(\beta_y) \mathbf{R}_z(\gamma_z), \quad (28)$$

extracts the anatomical joint rotations:

$$\alpha_x = \theta_{\text{nut}} = \text{atan2}(R_{21}, R_{22}) \quad [\text{Nutation } (> 0) / \text{Counternutation } (< 0) \text{ around } \hat{\mathbf{x}}_{\text{ML}}], \quad (29)$$

$$\beta_y = \theta_{\text{abd}} = \arcsin(-\text{clip}(R_{20}, -1.0, 1.0)) \quad [\text{Abduction} / \text{Out-flare } (> 0) \text{ around } \hat{\mathbf{y}}_{\text{AP}}], \quad (30)$$

$$\gamma_z = \theta_{\text{rot}} = \text{atan2}(R_{10}, R_{00}) \quad [\text{Axial spin} / \text{Torsion around } \hat{\mathbf{z}}_{\text{CC}}]. \quad (31)$$

Singularities at $\beta_y = \pm 90^\circ$ ($|R_{20}| \approx 1$) are handled via $\alpha_x = \text{atan2}(-R_{01}, R_{11})$ and $\gamma_z = 0$.

A.7 Articular contact centroid translation vector

The anatomical reference contact location $\mathbf{p}_c$ is defined as the centroid of the reference sacral articular facet ROI:

$$\mathbf{p}_c = \frac{1}{|\mathcal{R}_S|} \sum_{i \in \mathcal{R}_S} \mathbf{p}_{S,i}^0. \quad (32)$$

The relative displacement vector of this joint contact point in global space is:

$$\mathbf{d}_{\text{world}} = (\mathbf{R}_I - \mathbf{R}_S) \mathbf{p}_c + (\mathbf{t}_I - \mathbf{t}_S). \quad (33)$$

643 Projecting $\mathbf{d}_{\text{world}}$ into the anatomical pelvic axes yields the component translations:

$$644 \quad \mathbf{d}_{\text{pelvis}} = \mathbf{P}^T \mathbf{d}_{\text{world}} = \begin{bmatrix} d_{\text{ML}} \\ d_{\text{AP}} \\ d_{\text{CC}} \end{bmatrix}. \quad (34)$$

645 A.8 Bilateral kinematic aggregation and cohort summary metrics

646 Relative rotations and translations are evaluated independently for the left (L) and right (R)
647 joints. The 3D resultant norms are:

$$648 \quad \|\boldsymbol{\theta}_k\| = \sqrt{\alpha_{x,k}^2 + \beta_{y,k}^2 + \gamma_{z,k}^2}, \quad \|\mathbf{d}_k\| = \sqrt{d_{\text{ML},k}^2 + d_{\text{AP},k}^2 + d_{\text{CC},k}^2}, \quad k \in \{L, R\}. \quad (35)$$

649 To prevent lateral sign cancellation between symmetric left and right joints while capturing total
650 mechanical compliance, cohort-level summary metrics are defined as bilateral means of absolute
651 component values:

$$652 \quad \theta_{\text{mag}} = \frac{1}{2}(\|\boldsymbol{\theta}_L\| + \|\boldsymbol{\theta}_R\|), \quad d_{\text{mag}} = \frac{1}{2}(\|\mathbf{d}_L\| + \|\mathbf{d}_R\|), \quad (36)$$

$$653 \quad \theta_{\text{nut}} = \frac{1}{2}(|\alpha_{x,L}| + |\alpha_{x,R}|), \quad |d_{\text{ML}}| = \frac{1}{2}(|d_{\text{ML},L}| + |d_{\text{ML},R}|), \quad (37)$$

$$654 \quad |d_{\text{AP}}| = \frac{1}{2}(|d_{\text{AP},L}| + |d_{\text{AP},R}|), \quad |d_{\text{CC}}| = \frac{1}{2}(|d_{\text{CC},L}| + |d_{\text{CC},R}|). \quad (38)$$

655 Bilateral translation asymmetry is defined as the absolute magnitude difference:

$$656 \quad \Delta_{\text{asym}} = \left| \|\mathbf{d}_L\| - \|\mathbf{d}_R\| \right|. \quad (39)$$

657 A.9 Morphometric triad definitions

658 The three structural covariates comprising the morphometric triad are evaluated directly from
659 landmark coordinates warped to each subject's geometry:

- 660 1. **Anteroposterior Inlet Diameter (d_{AP}):** Conjugate vera, defined as the Euclidean
661 distance between the anterior sacral promontory ($\mathbf{p}_{\text{prom}}$) and the superior/posterior pubic
662 symphyseal crest ($\mathbf{p}_{\text{pubis}}$):

$$663 \quad d_{\text{AP}} = \|\mathbf{p}_{\text{prom}} - \mathbf{p}_{\text{pubis}}\|. \quad (40)$$

- 664 2. **Biischadic Width (w_{biisch}):** Interspinous outlet diameter, defined as the transverse

distance between the left and right ischial spines ($\mathbf{p}_{\text{spine},L}$ and $\mathbf{p}_{\text{spine},R}$):

$$w_{\text{biisch}} = \|\mathbf{p}_{\text{spine},L} - \mathbf{p}_{\text{spine},R}\|. \quad (41)$$

3. Subpubic Arch Angle (α_{subpubic}): Planar angle measured at the inferior apex of the pubic symphysis ($\mathbf{p}_{\text{apex}}$) between vectors directed along the descending left and right inferior pubic/ischial rami ($\mathbf{p}_{\text{rami},L}$ and $\mathbf{p}_{\text{rami},R}$):

$$\cos \alpha_{\text{subpubic}} = \frac{\mathbf{v}_L \cdot \mathbf{v}_R}{\|\mathbf{v}_L\| \|\mathbf{v}_R\|}, \quad \text{where } \mathbf{v}_L = \mathbf{p}_{\text{rami},L} - \mathbf{p}_{\text{apex}}, \quad \mathbf{v}_R = \mathbf{p}_{\text{rami},R} - \mathbf{p}_{\text{apex}}. \quad (42)$$

Total pelvic bone volume (Vol) is evaluated via direct spatial integration over tetrahedral bone elements: $\text{Vol} = \sum_{e \in \Omega_{\text{bone}}} V_e$.

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Author contributions

CRedit authorship contribution statement. Petr Henyš: Conceptualization; Methodology; Software; Formal analysis; Investigation; Data curation; Visualization; Funding acquisition; Writing – original draft; Writing – review & editing. Niels Hammer: Supervision; Writing – review & editing. All authors reviewed and approved the final manuscript.

Conflict of interest statement

The authors declare no conflicts of interest.

Supporting information

Supporting Information includes Supplementary Methods (implicit mapping, load pullback, SIJ kinematics, and size-scaling model details) and Supplementary Tables S1–S4.

Data Availability Statement

The derived data supporting the findings of this study (summary tables and figures) are provided with the manuscript. Subject-level derived datasets and analysis outputs are available in a blinded repository for peer review and will be made publicly available upon publication.

Code Availability Statement

All custom code used to post-process the finite-element outputs and perform the statistical analyses is available in the same blinded repository for peer review and will be made publicly available upon publication.

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